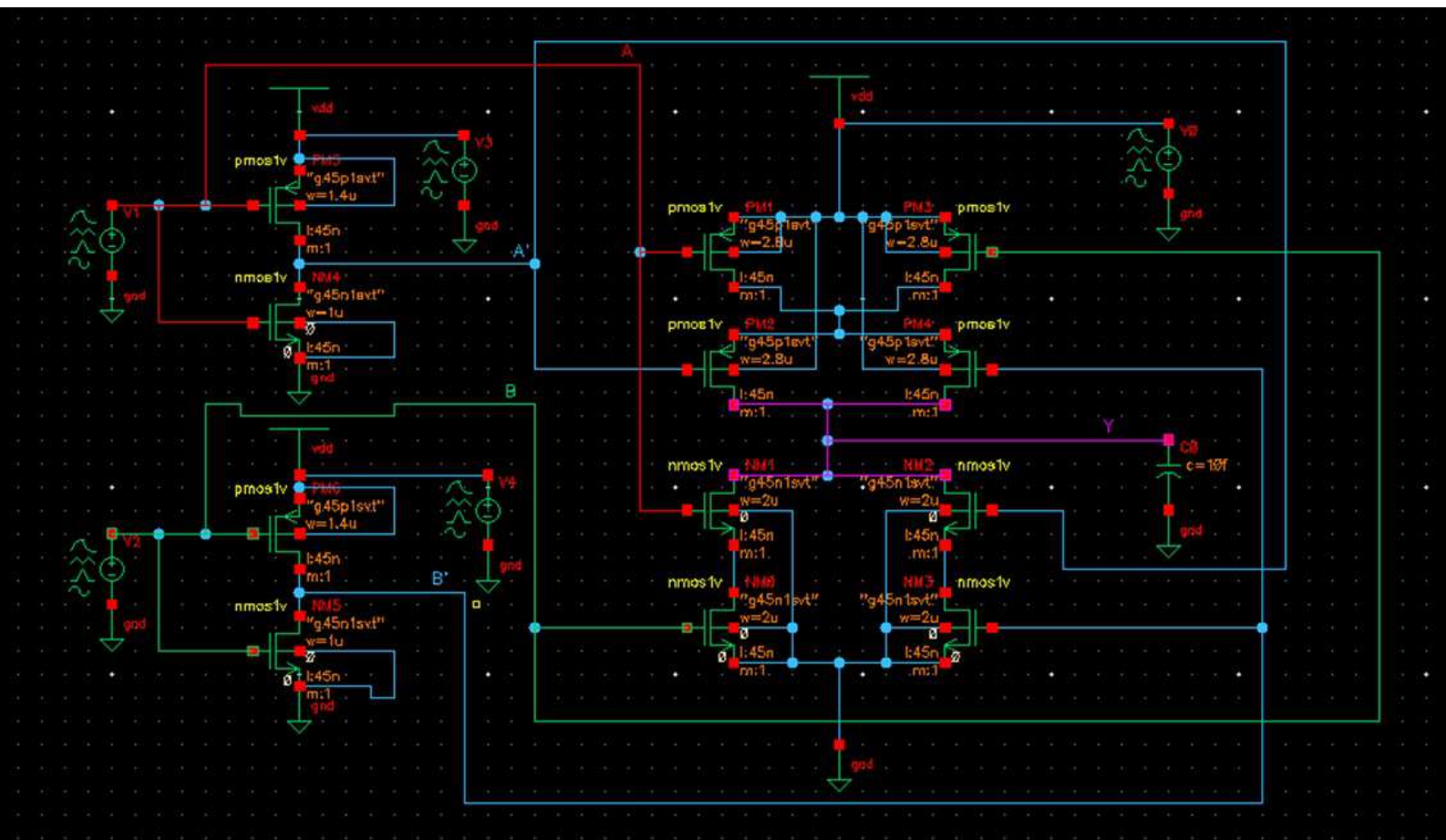
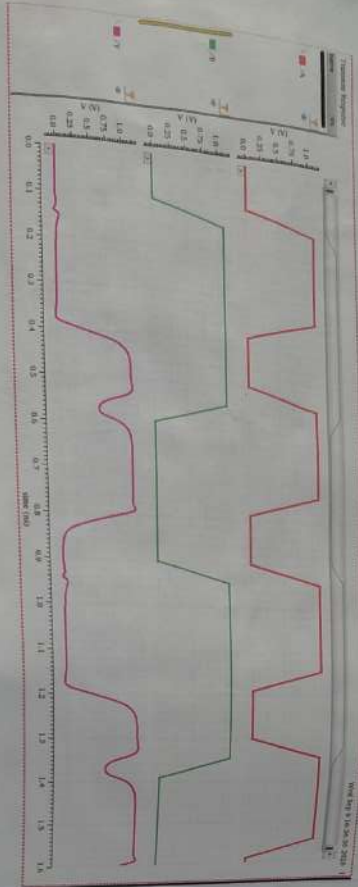






A=0, B=20, C=0, #10
A=20, B=20, C=20, #10
A=20, B=20, C=20, #10

Experiment Name :

NMOS transistor : An NMOS (N-channel MOSFET) is

- usually used as a pull-down transistor.
- It turns ON when the gate voltage is HIGH.
- It converts the output to the ground when ON.
- It forms the pull-down network (PDN).

Therefore, $V_G = V_{DD} \Rightarrow$ NMOS on.
 $V_G = 0 \Rightarrow$ NMOS off.

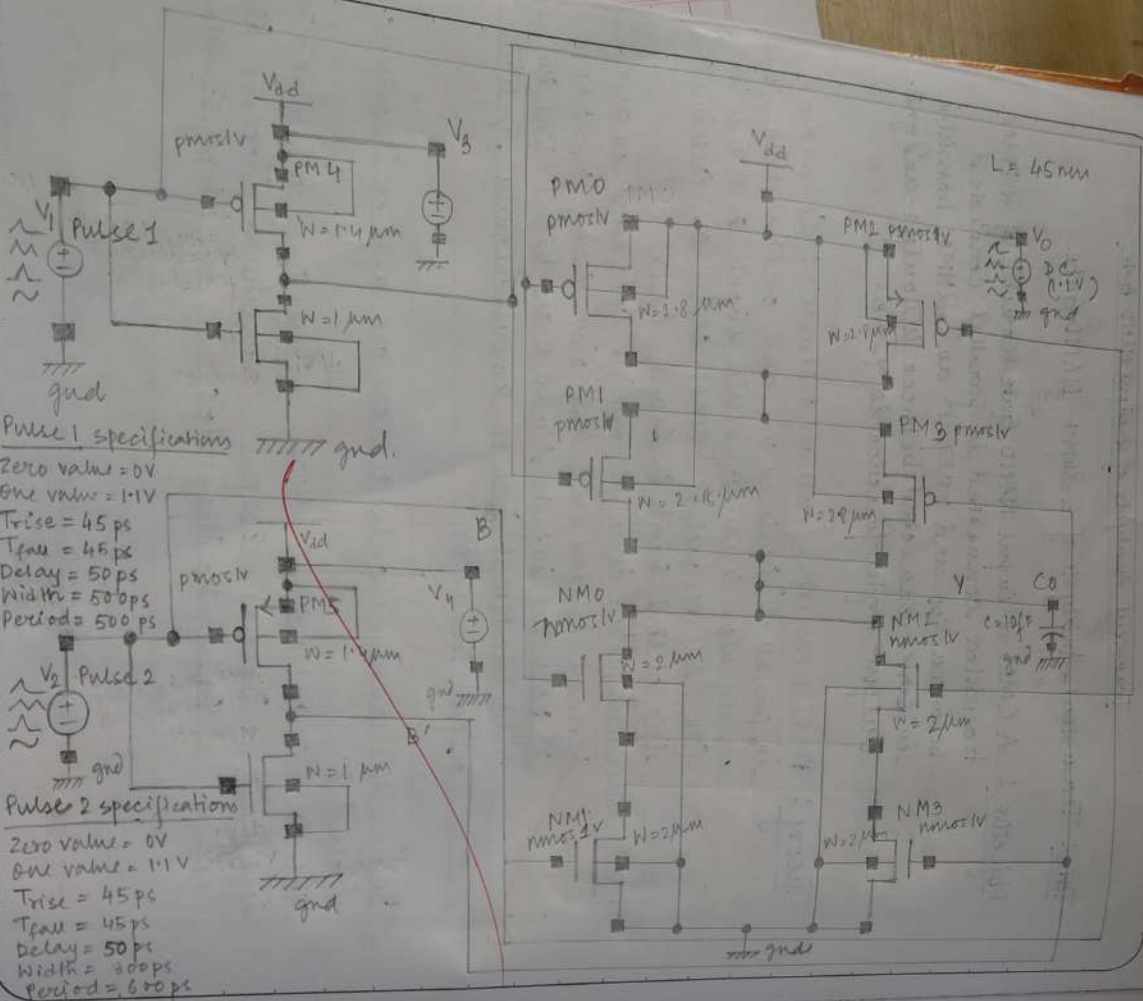
Transient analysis : In this experiment, pulse signals are applied to the inputs A and B. The transient analysis allows us to observe:

- Input waveforms A and B.
- Output waveform Y.
- Logic transitions.
- Propagation delay.
- Rise and fall time.

For pulse 1, transient analysis shows the following:

Pulse 1:	Pulse 2:
Zero value = 0V	Zero value = 0V
One value = 1.1V	One value = 1.1V
Trise = 45ps	Trise = 45ps
Tfall = 45ps	Tfall = 45ps
Delay = 100ps	Delay = 100ps
Width = 500ps	Width = 500ps
Period = 500ps	Period = 600ps

Teacher's Signature



Experiment Name: Transient analysis of 2-input CMOS XOR gate

Title: Transient analysis of 2-input CMOS XOR gate using PMOS and NMOS transistors

Objective: To design and simulate a 2-input CMOS XOR gate using PMOS and NMOS transistors and to perform transient analysis by applying two pulse inputs A and B and observing the output waveform.

Theory: The circuit consists of PMOS and NMOS transistors. The input signals are A and B and the output is Y.

For an XOR gate, $Y = A \oplus B$.

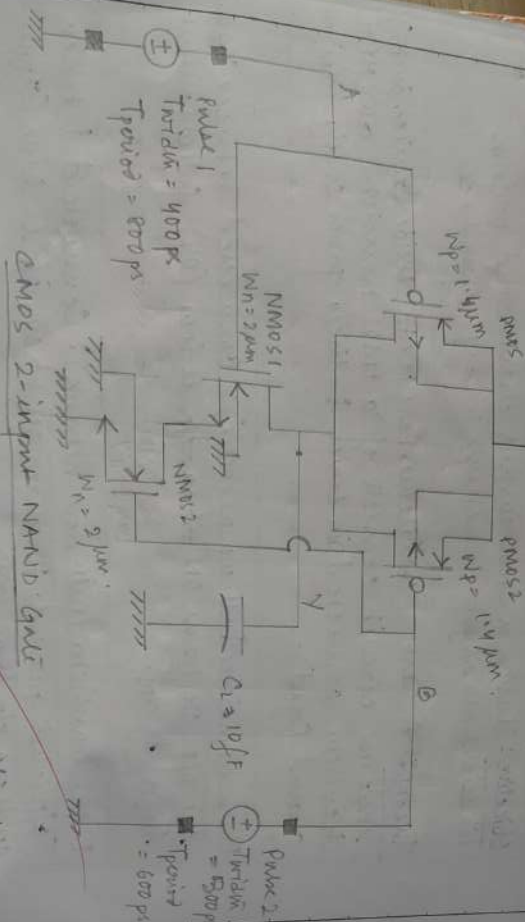
Truth Table: A B Y = A ⊕ B

0	0	0
0	1	1
1	0	1
1	1	0

The inverter in the circuit generates the complement signals A' and B' required by CMOS.

PMOS Transistor: A PMOS (p-channel MOSFET) is used as a pull-up transistor in CMOS. It turns ON when the gate voltage is low (0V) and connects the output to VDD. It is used in the pull-up network. Source is generally connected to VDD.

Therefore, $V_{G1} = 0 \Rightarrow$ PMOS ON
 $V_{G1} = V_{DD} \Rightarrow$ PMOS OFF



Pulse 1 specification

Zero value = 0V
 High value = 1.1V
 $T_{rise} = 45 \text{ ps}$
 $T_{fall} = 45 \text{ ps}$
 Delay = 50 ps
 Width = 400 ps
 Period = 800 ps

Pulse 2 specification

Zero value = 0V
 High value = 1.1V
 $T_{rise} = 45 \text{ ps}$
 $T_{fall} = 45 \text{ ps}$
 Delay = 50 ps
 Width = 300 ps
 Period = 600 ps

Experiment Name: Transient analysis of 2-input NAND Gate

Page No.

Title: Transient analysis of 2-input NAND Gate

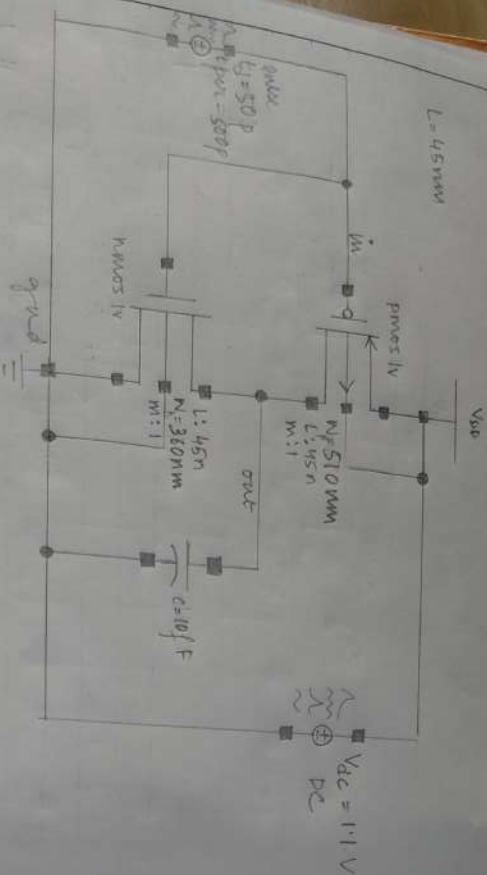
Objective: A CMOS 2-input NAND gate consists of 2 PMOS transistors connected in parallel connected between V_{DD} and output and 2 NMOS transistors in series connected between the output and ground. Output logic function is $Y = AB$.

Theory: PMOS transistors (Pull-up Network): Connected in parallel between V_{DD} and output with $W_p = 4 \mu\text{m}$. If either input A or B is low (0V), the respective PMOS will be ON, pulling the output high to V_{DD} (1.1V).
 NMOS transistors (Pull-down Network): Connected in series between output and ground with $W_n = 2 \mu\text{m}$. If both the inputs A and B are high (1.1V), the NMOS will be ON, pulling the output low to 0V.

In transient analysis, pulse sources are applied to inputs A and B.

Pulse 1: Pulse width = 400 ps, $T_{period} = 800 \text{ ps}$, $T_{propagation\ delay} = 50 \text{ ps}$
Pulse 2: Pulse width = 300 ps, $T_{period} = 600 \text{ ps}$, $T_{propagation\ delay} = 50 \text{ ps}$

Teacher's Signature



Pulse Specification

Zero value = 0V

Low value = 1.1V

Trise = 45ps

Tfaw = 45ps

Delay = 50ps

Period = 50ps

CMOS INVERTER

Experiment Name: Transient Analysis of 2 input NOT gate

Title: To study the transient and DC analysis of 2 input CMOS NOT gate in 45nm technology.

Objective: Design a CMOS inverter circuit and perform transient analysis to measure its key performance.

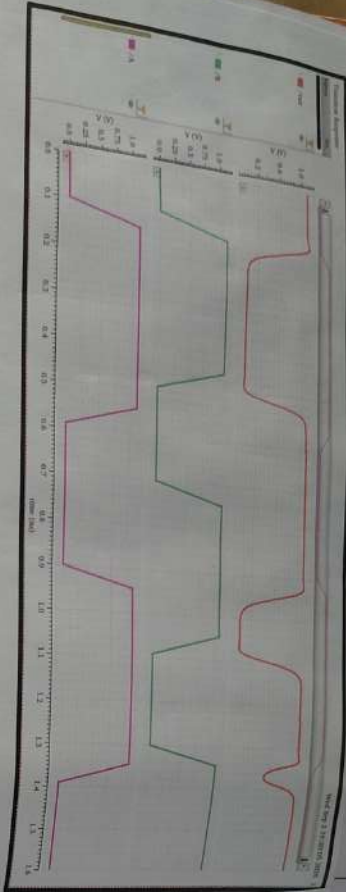
Theory: A CMOS (Complementary Metal Oxide Semiconductor) inverter circuit consists of a PMOS transistor and NMOS transistor connected in series between power supply (V_{DD}) and ground (GND).

PMOS transistor (Pull-up network): Connected between V_{DD} and the output. It turns ON when the input is low (0V), pulling output high to V_{DD} .

NMOS transistor (Pull-down network): Connected between the output and ground. It turns ON when the input is high (V_{DD}), pulling output low to 0V.

Transient analysis: In transient analysis, we measure the propagation delay between the input and output, the rise and fall delay (time taken for the signal to transfer between 10% and 90% of its peak). In DC analysis, by investigating the VTC, one can find the threshold voltage (V_{th}) defining noise margin.

$$V_{DD} = 1.1V, W_p = 510nm, W_n = 310nm, C_L = 10fF$$



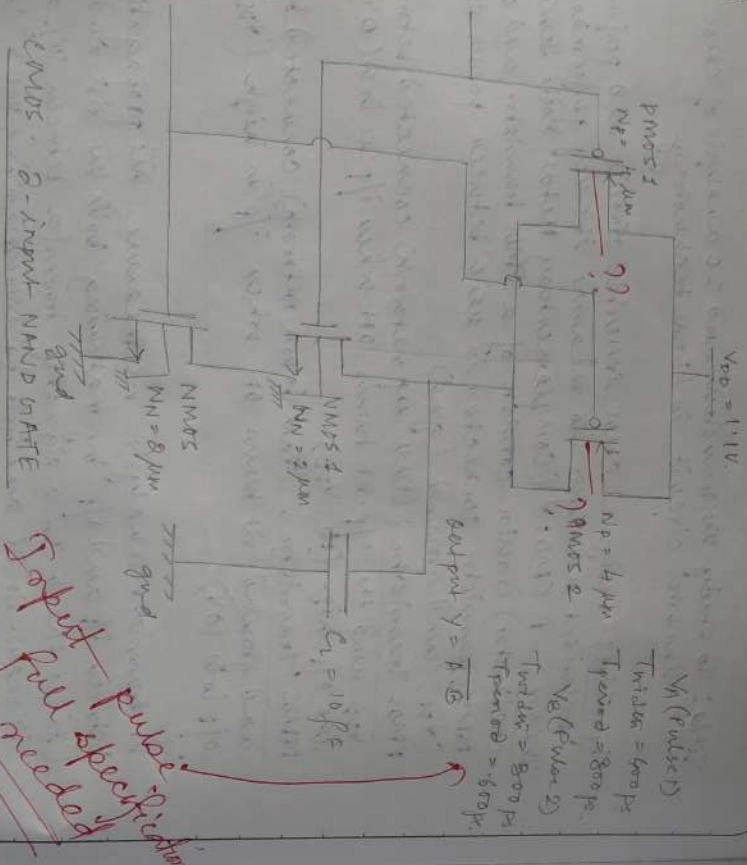
Handwritten signature

Handwritten notes:
 Length of wire used = 1.0 m
 Area of cross-section = 1.0 mm²
 Resistance = 1.0 Ω
 Voltage = 1.0 V
 Current = 1.0 A
 Power = 1.0 W
 Heat = 1.0 J
 Time = 1.0 s
 Temperature = 1.0 °C

Experiment Name :

Page No. _____
 Date: _____

Teacher's Signature



Logic Symbol of NAND gate

A B Y

Truth Table of NAND gate

1 0 1
1 0 1
0 1 1
0 1 1
1 1 0
1 1 0

Experiment Name:

Transient Analysis of 2-input NAND gate

Title: To study the transient and the analysis of 2-input CMOS NAND gate in 45 nm technology.

Objective: A CMOS 2-input NAND gate, connected as a CMOS transistors in parallel connected between V_{DD} and output and 2 NMOS transistors in series connected between output and ground. Output logic function is $Y = A \cdot B$.

PNOS techniques (pull up network): Connected in parallel between V_{DD} and output with sizing $W_p = 4 \mu m$. If either input A or B is low (0 V), the respective PMOS forms ON, pulling the output high is V_{DD} (1.1 V).

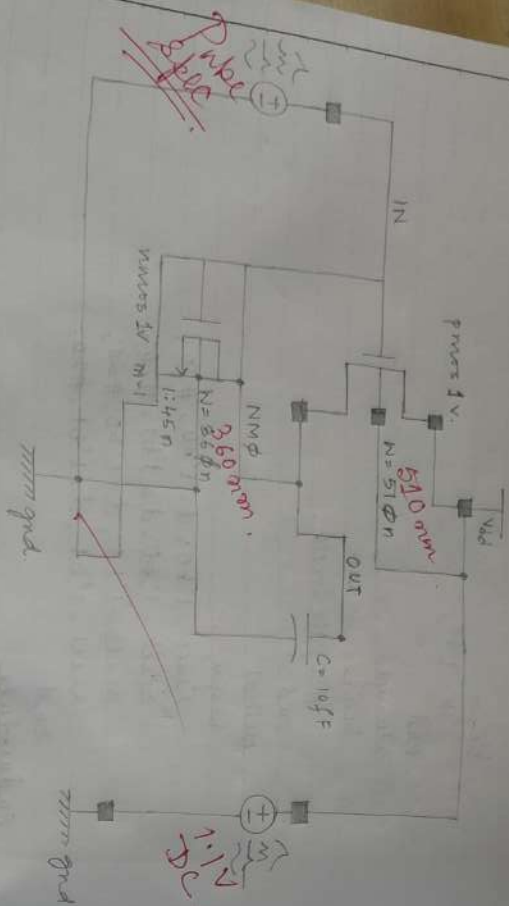
NMOS techniques (pull down network): Connected in series between output and ground with sizing $W_n = 8 \mu m$, only when both inputs A and B are high (1.1 V) both NMOS turn ON simultaneously, pulling the output low to 0 V.

In transient analysis, pulse sources are applied to inputs A and B:

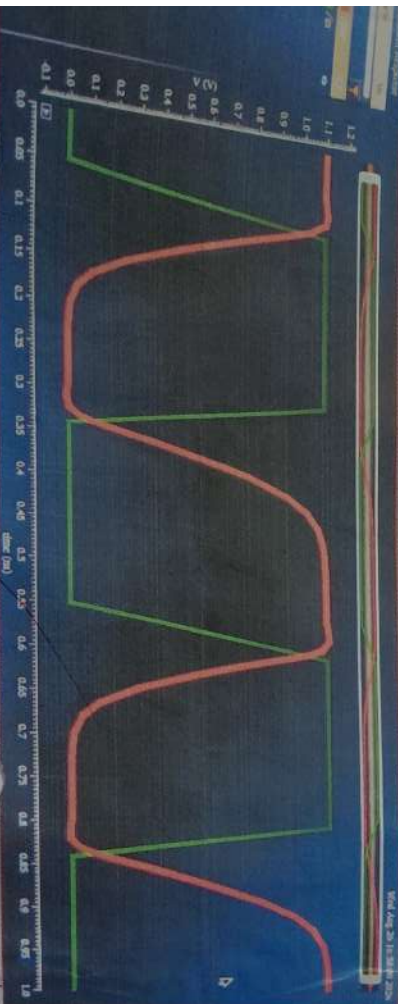
Input A (Pulse 1): Pulse width = 400 ps
 $T_{period} = 800$ ps.

Input B (Pulse 2): Pulse width = 800 ps.
 $T_{period} = 600$ ps.

Teacher's Signature



CMOS Inverter



Experiment Name:

Page No.
Date:

Title: To study the transient and DC analysis of CMOS inverter circuit in ESW technology.

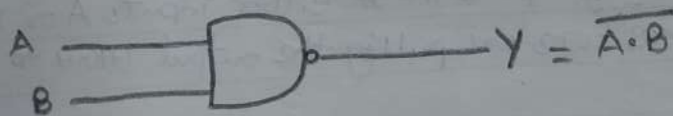
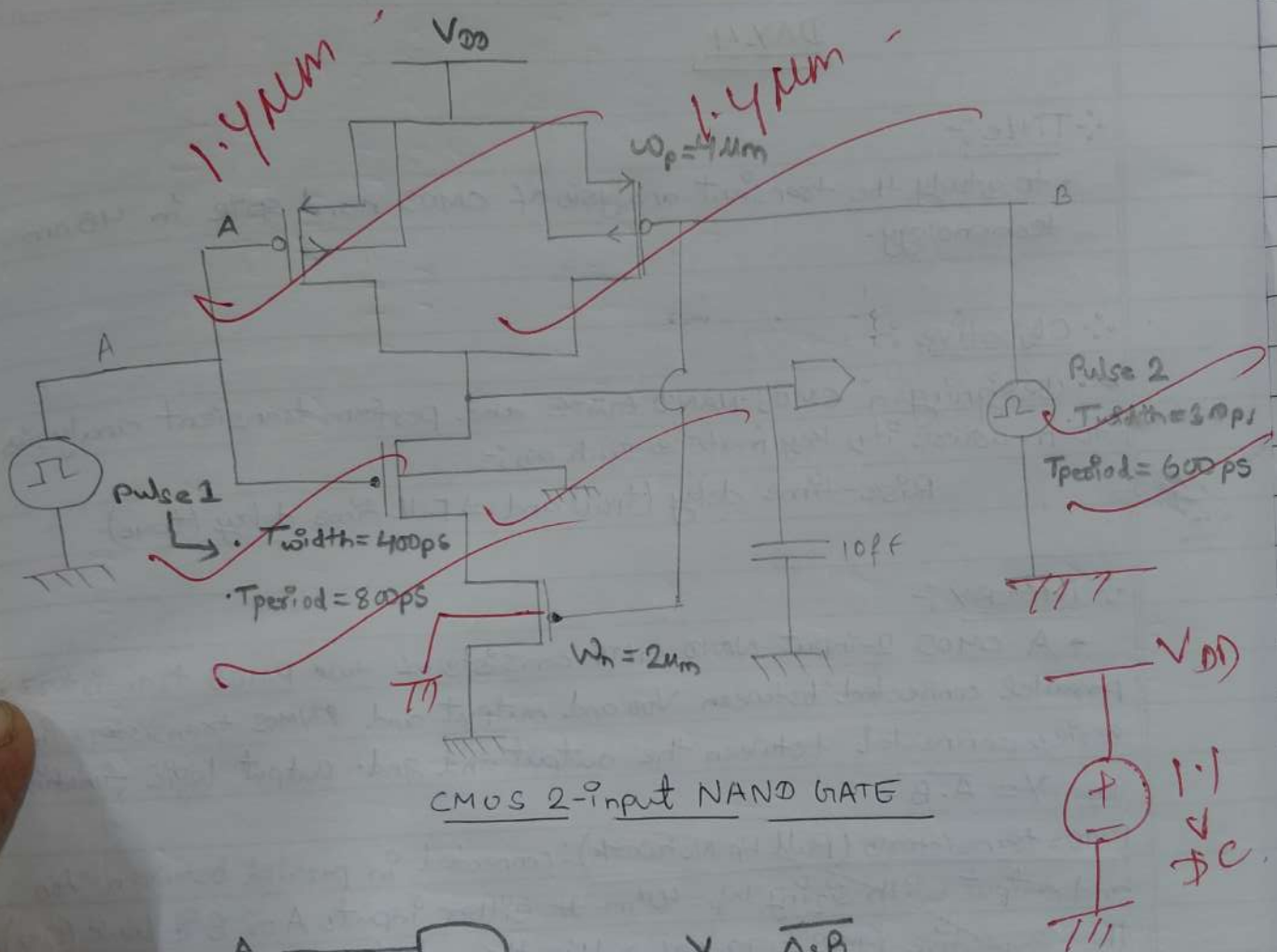
Objective: Design a CMOS inverter circuit and perform transient analysis to measure its key performance metrics. A CMOS (Complementary Metal Oxide Semiconductor) inverter consists of a PMOS transistor and an NMOS transistor connected in series between power supply (VDD) and ground (GND). The PMOS transistor is connected between VDD and the output node, and the NMOS transistor is connected between the output node and ground. The input signal is applied to the gates of both transistors. The output signal is taken from the common drain connection of both transistors. The PMOS transistor is connected to VDD and the NMOS transistor is connected to ground. The output signal is taken from the common drain connection of both transistors. The input signal is applied to the gates of both transistors. The output signal is taken from the common drain connection of both transistors.

In transient analysis, we measure the propagation delay between input and output by matching both the rise and fall delay (time taken for a signal to transfer between 10% and 90% of its peak). In DC analysis by investigating the VTC, we establish the threshold voltage (Vth) defining noise margin.

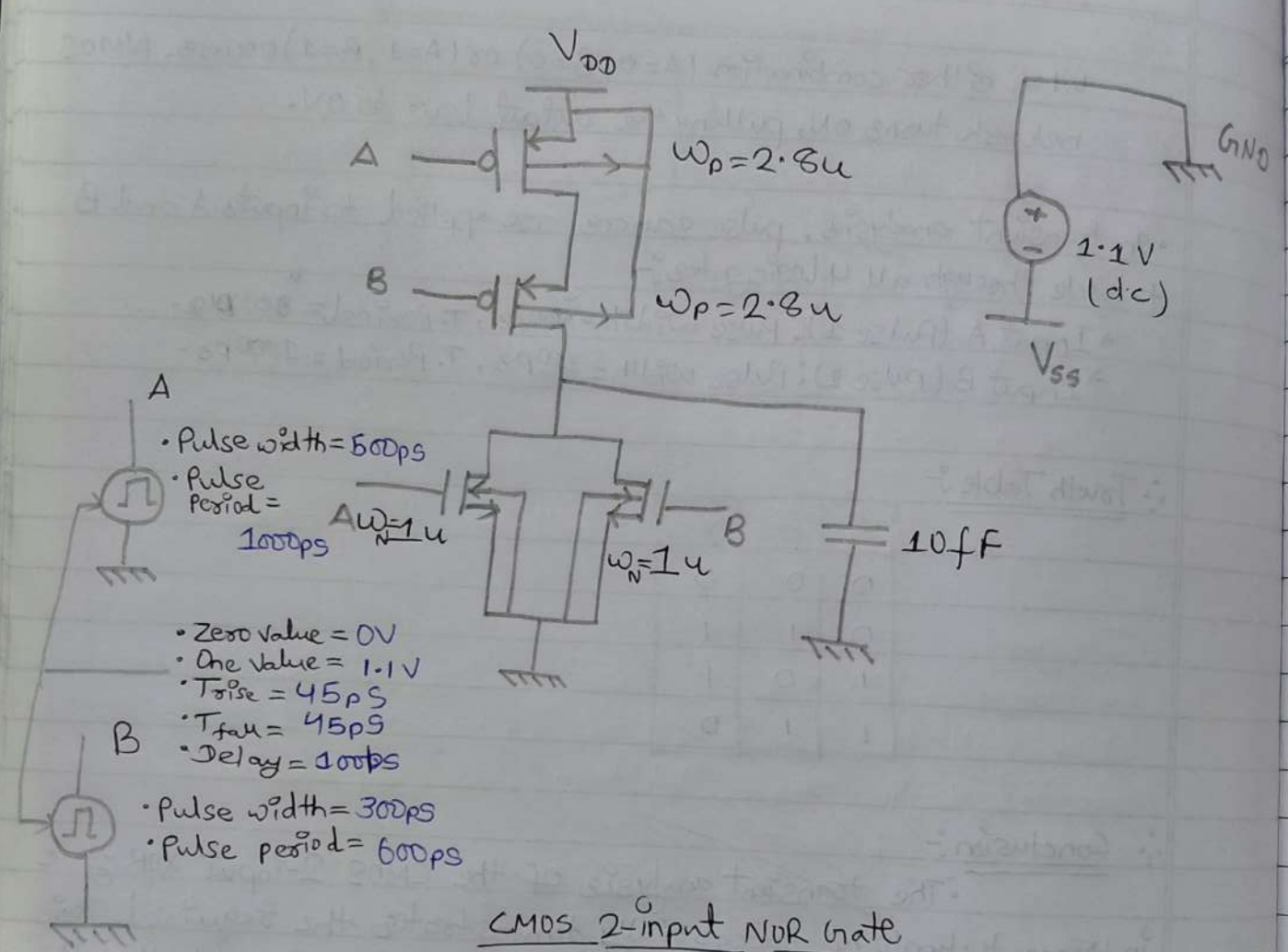
Observation: Transient analysis: VDD=1.1V, NM=120nm, NP=170nm, C=17.5pF

Rise delay (trise) = 26.89 ps
Fall delay (tfall) = 26.80 ps
Average delay = 26.69 + 26.80 = 26.745 ps

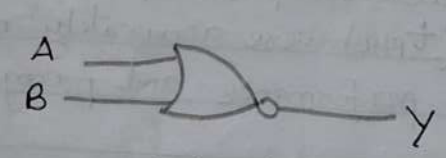
Teacher's Signature:



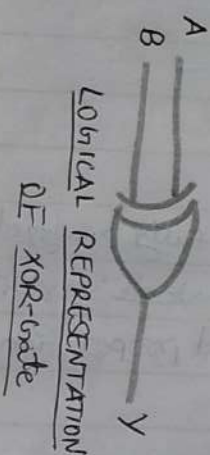
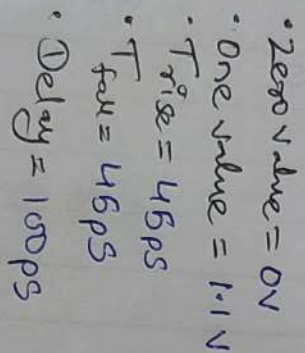
Logic Symbol of NAND GATE.



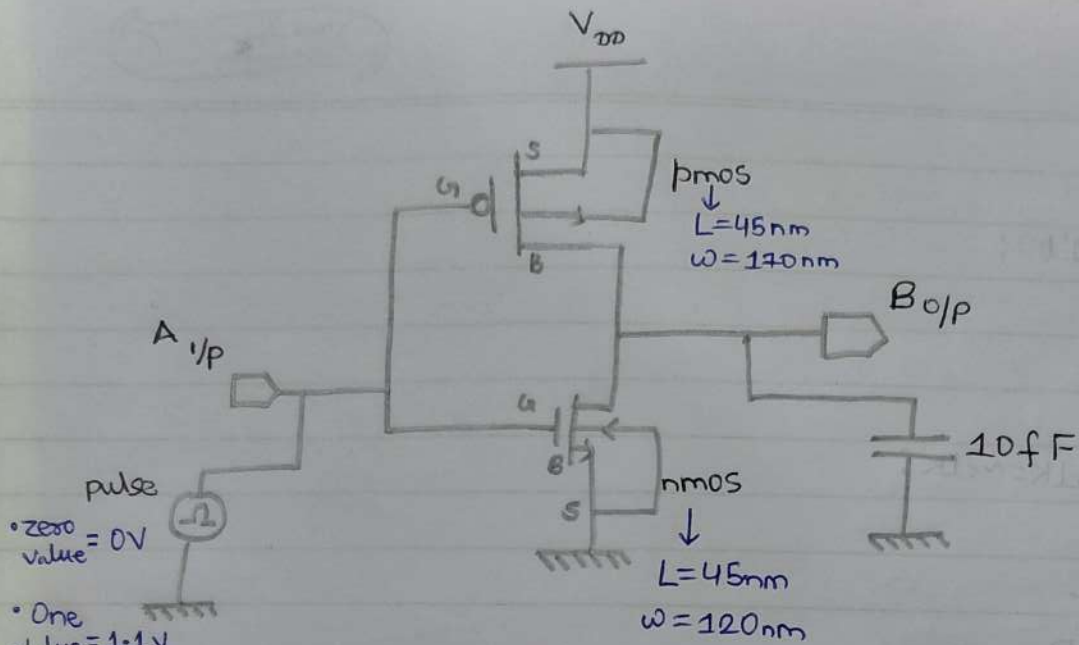
CMOS 2-input NOR gate



Logical Diagram



CMOS 2-input XOR GATE



- zero value = 0V
- One value = 1.1V
- $T_{\text{rise}} = 45\text{ps}$
- $T_{\text{fall}} = 45\text{ps}$
- Delay = 100ps
- width = 200ps
- Period = 400ps

$$\frac{W_{\text{pmos}}}{W_{\text{nmos}}} = 1.4$$

$$W_{\text{nmos}} = 360$$

$$W_{\text{nmos}} = 120\text{nm}$$

$$W_{\text{pmos}} = 570$$

$$W_{\text{pmos}} = 170\text{nm}$$

Good